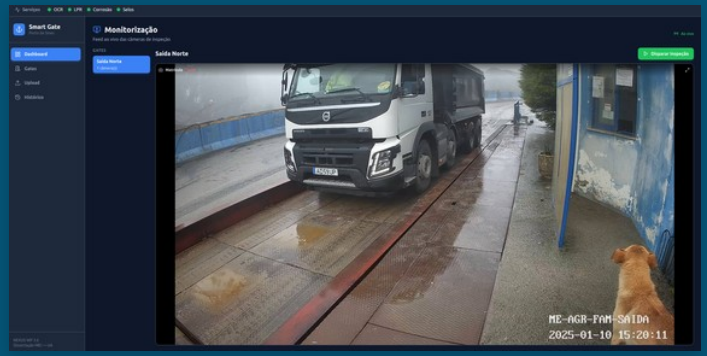
NEXUS · Port Scene Analytics (WP 3.6)

SMART GATE

One inspection for plate, container code, seal, and corrosion at the port gate





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Port gates are inspected by hand today

For every truck that passes, four checks are made:

1. License plate
2. Container code (ISO 6346)
3. Seal
4. Corrosion

One truck at a time

Each check is done separately, as the trucks go through.

No shared record

Nothing links the four checks to the same truck, gate, and moment.

The NEXUS project

NEXUS is a national innovation agenda for the green and digital transition of transport, logistics, and mobility, funded by Portugal's Recovery and Resilience Plan.

35

partners in the consortium

Sines

led by the Port of Sines Authority

WP 3.6

Port Scene Analytics, in the Smart Gates & Terminals pillar

Within WP 3.6, IEETA developed four computer vision models for the gate. Smart Gate brings them together in one platform.

The four technologies



Mod. 1 [1]

Container code

Reads the ISO 6346 code



Mod. 2 [2]

Corrosion

Estimates corrosion from the image



Mod. 3 [3]

License plate

Reads the truck's plate automatically



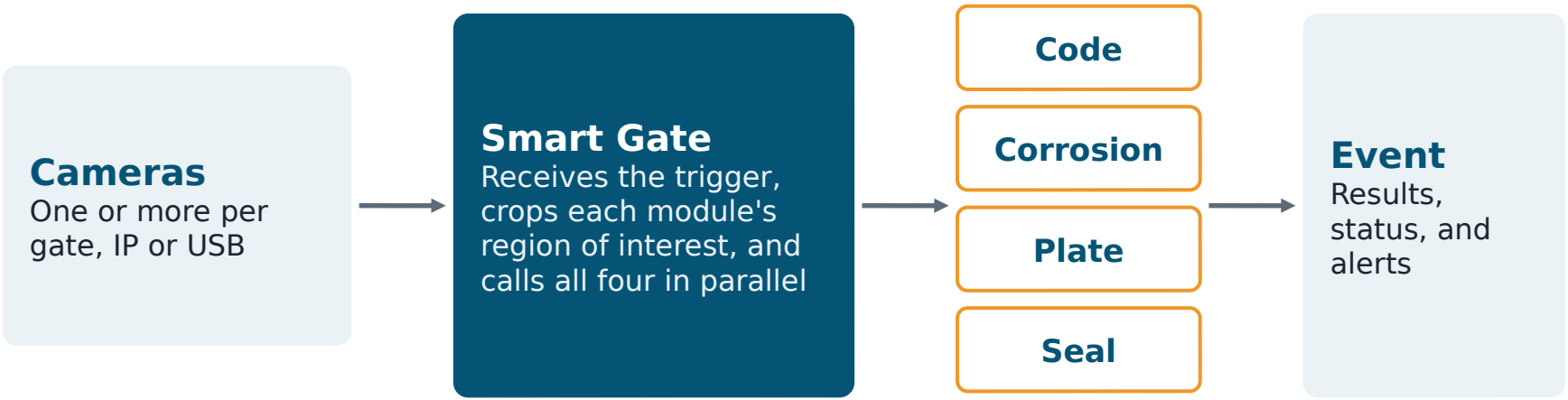
Mod. 4

Seal detection

Checks seal presence

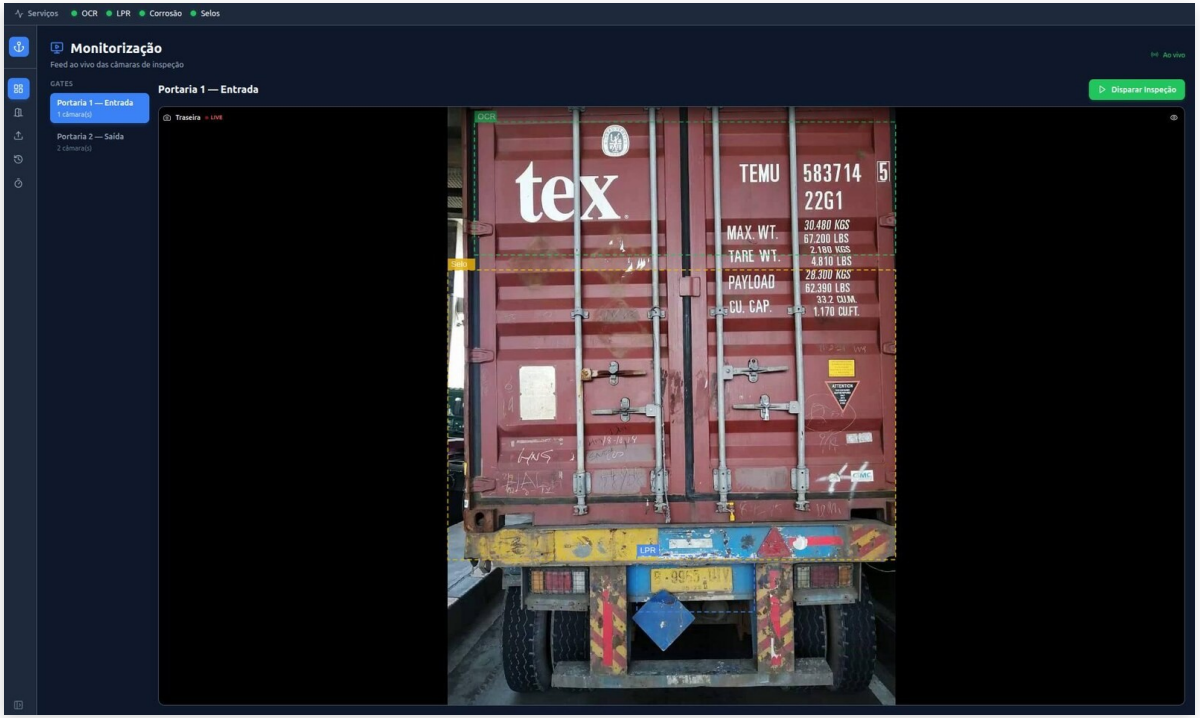
One Inspection Event

These four technologies don't share a language: different response times, different formats, some instant, some take seconds. Smart Gate runs them in parallel and merges their results.



All tied to the same truck, gate, and moment. The full system is deployed with Docker Compose on a GPU server.

Live video from each gate



Real-time camera stream, as seen by the operator before an inspection fires.

Each module's region of interest is drawn as a dashed line over the image.

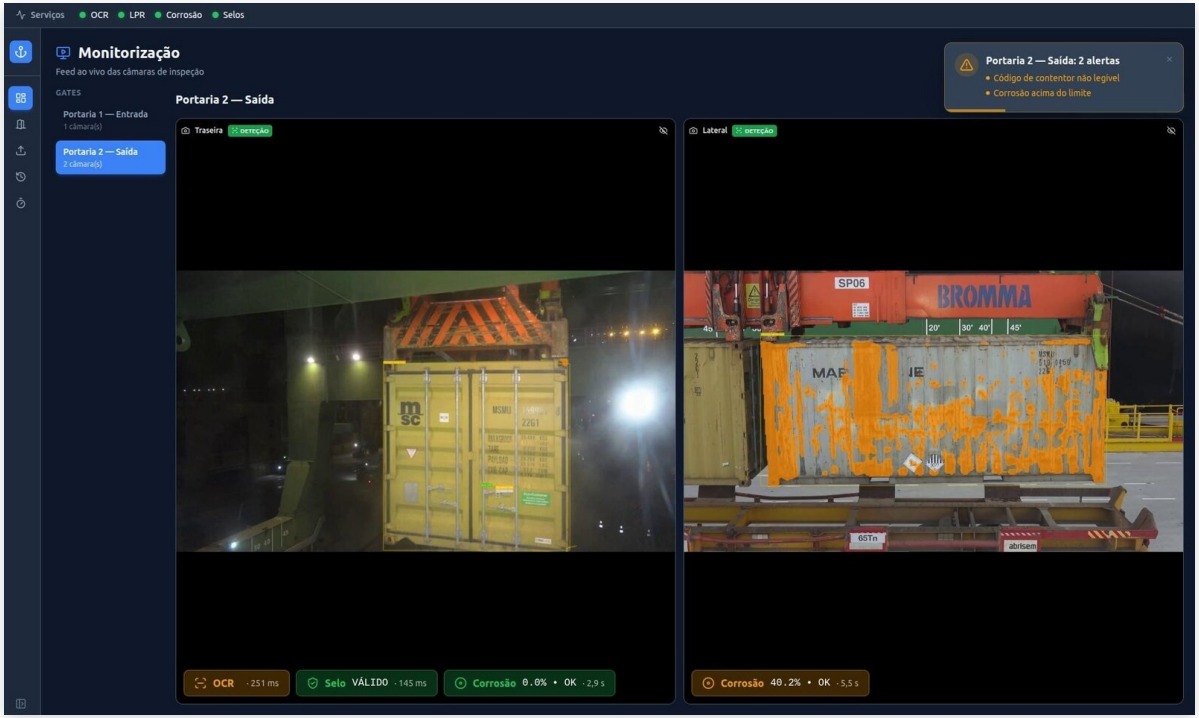
The stream reconnects automatically if the camera drops.

Results and alerts in one view

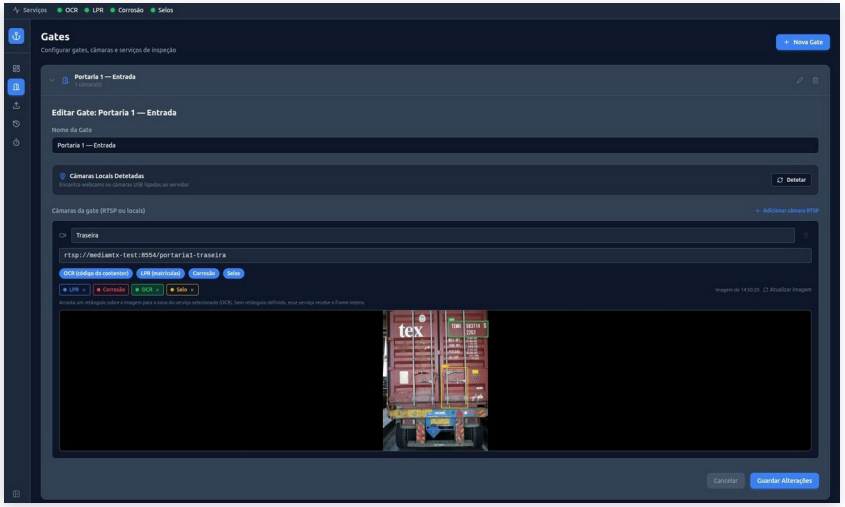
Annotated images from every camera at the gate.

One card per module, with its result and response time.

Alerts when something is wrong: here, an unreadable container code and corrosion above the threshold.

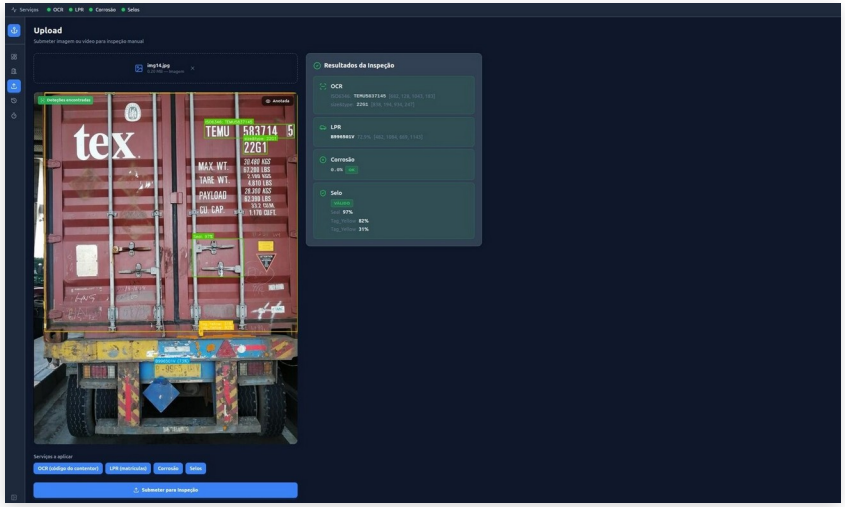


Gate setup and file upload



Gates

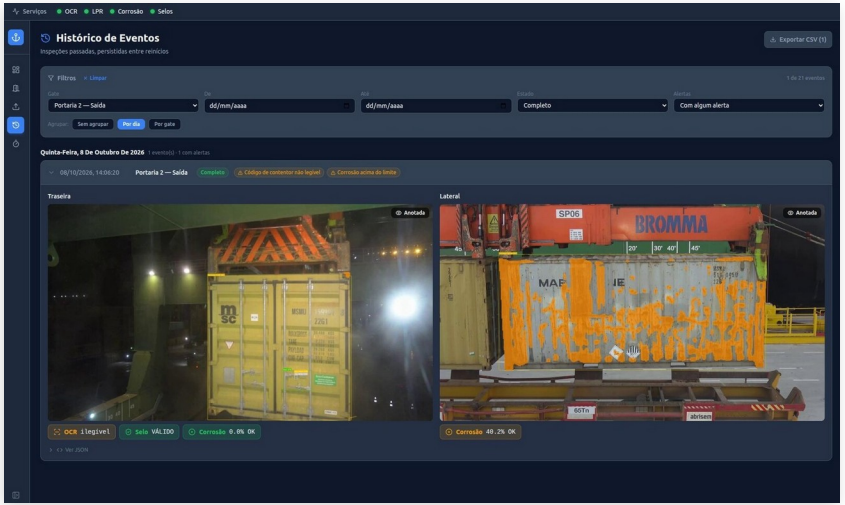
Cameras, the modules assigned to each camera, and one region of interest per module.



Upload

An image or a recorded video, analyzed by the four modules.

History and response time



History

Stored inspections, filtered by gate, day, status, and alerts, with CSV export.



Latency

Median and 95th percentile per module, and changes across inspections.

Tested end to end

4–5 s

median time for a full inspection with all four modules and one camera

7.5 s

with two cameras at the same gate; about 15 s with four

3

rounds of tests: response time, several cameras per gate, and failures

100%

of the failure scenarios tested still produced an inspection event

Laboratory tests on a GPU server, with emulated cameras streaming real port images.

Why Smart Gate

1

One record per truck

Plate, container code, seal, and corrosion in a single inspection event, with alerts.

2

Keeps working when something fails

If a module or a camera fails, the others still answer and the event shows what is missing.

3

Built for real gates

Several cameras per gate, a region of interest per module, live video, and a full history.

4

Easy to deploy

The whole system, with the four models, starts with one command on a single GPU server.

Thank you

SMART GATE

Four AI models, one platform, one inspection event per truck.

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References

[1] J. Santos, D. Canedo, A. J. R. Neves, “Automating Code Recognition for Cargo Containers,” Electronics, vol. 14, no. 22, art. 4437, 2025. doi.org/10.3390/electronics14224437

[2] D. Ornelas, D. Canedo, A. J. R. Neves, “Semantic Segmentation of Corrosion in Cargo Containers Using Deep Learning,” Sustainability, vol. 17, no. 14, art. 6480, 2025. doi.org/10.3390/su17146480

[3] D. Ornelas, J. Santos, D. Canedo, A. J. R. Neves, “License Plate Recognition in Port Scene Analytics,” RECPAD 2025, Aveiro, Portugal, pp. 159–160.